

Policy AI Lab Product Requirements

Document	Product Requirements
Solution	Policy AI Lab
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This document defines the product requirements for Policy AI Lab, an internal decision-support workspace that helps MOMAH teams develop new Regulations, Requirements and Rules from the identification of a legislative need through policy research, benchmarking, impact assessment, option decision, drafting, legal readiness and consultation feedback, ending in a Review-Ready Regulatory Draft and a Decision Package. It states what the product must do, the principles that govern how AI assistance and human decisions interact, the quality and operating characteristics expected of the solution, the data and system boundaries within which it operates, and the decisions that remain open. Requirements are grouped by module; each carries an identifier, a priority, a delivery phase and a source. Priority uses P0 must have, P1 should have, P2 nice to have, P3 backlog. Phase indicates the earliest phase in which the requirement is expected in working form; the Prototype demonstrates requirements with predefined states unless stated otherwise.

1 Sources and conventions

The requirements in this document derive primarily from the alignment sessions held with the Solution Owner between 11 August and 2 September 2026, in which the business process, roles, methodology and scope were worked through step by step, and from MOMAH's Legislative Development Methodology that those sessions applied. The business baseline confirmed with MOMAH in April 2026 is the antecedent of this work; where a Solution Owner decision differs from it, the decision governs, and Section 10 records where the baseline was confirmed, refined or deferred.

Reference	Source	Used for

M-0811 to M-0902	Solution Owner alignment sessions of 11, 16, 17, 18, 19, 24, 26 and 31 August and 2 September 2026	Primary source. Business process and sequence, role responsibilities, scope decisions, AI and human-confirmation principles, Benchmarking and Impact Assessment rules, knowledge-base and data policies. Cited by date in the Source column.
LDM	MOMAH Legislative Development Methodology, March 2026, and the Solution Owner's alignment note of 17 May 2026	Nine official steps, three phases, nine Policy Study contents, 39-item checklist.
BM / IA	Policy Benchmarking Methodology and Impact Assessment Methodology (ByteDance, August 2026, endorsed in sessions M-0818 and M-0819, external expert review September 2026)	Detailed rules behind the Benchmarking and Impact Assessment modules.
DSR	Data and integration support request, 20 July 2026	Platform integrations and DataBank coverage.
BB	Business baseline confirmed with MOMAH during the Preparation phase, April 2026	Antecedent. Capability list BB-01 to BB-15, quality attributes BQ-01 to BQ-06, scope and exclusions, personas, integrations and data sources. Reconciled in Section 10.

Requirement identifiers use the form PR-<module>-<number>. Session references (M-date) identify the Solution Owner decision that shaped a requirement; BB-nn and BQ-nn identify the baseline capability or quality attribute it refines, as listed in Section 10. The Source column names the baseline capability or scope item the requirement derives from, and the session date where a Solution Owner decision shaped it. Open decisions that affect a requirement are listed in Section 9 as OD-n. Section 10 records how each baseline capability is carried, refined or deferred.

2 Product overview

Positioning. Policy AI Lab was first framed as a decision-support platform that moves MOMAH's policy lifecycle from a manual, document-centric process to an evidence-based, AI-augmented

one. Through the Solution Owner sessions the product is positioned as a dynamic policy analysis engine for developing new Regulations, Requirements and Rules: register a legislative need, clarify the question, research the evidence, benchmark international mechanisms, assess impact, compare options, draft, check legal conflicts and readiness, handle consultation feedback and assemble a Decision Package. AI assists at every step; responsible people make every decision.

Sequence. The baseline lifecycle orchestration (BB-13) runs draft, simulation, review and approval. The Solution Owner confirmed on 11 August that the sequence follows the Legislative Development Methodology: research every aspect first, including benchmarking and impact assessment, and draft only after the preferred option is decided. The baseline capabilities are retained; their order changes.

Lifecycle coverage. Against the nine steps of the methodology, Policy AI Lab is the working environment for steps 1 to 4 and 6 (identify cases, develop the initial concept, checklist review, refine the draft, update the draft after feedback). Step 5, publication on the Istitlaa platform, is performed by responsible users outside the solution with material exchanged manually. Steps 7 to 9 (publication, circulation and archiving, implementation and measurement) remain outside the solution; step 9 may trigger a new case at step 1.

Delivery phases and instrument scope. The current delivery covers drafting new instruments from scratch. The baseline capability for amendments to existing policy (BB scope) is deferred to a later phase and shown in the workspace as a labelled placeholder (M-0826). Outputs are Regulation, Requirement and Rule drafts; Guides and Laws are held in the knowledge base as references but are not produced (M-0817). Draft version iteration and the full Stakeholder Review flow remain inside Policy AI Lab and are not handed off to Sanad (M-0831). Statutory approval, electronic signature and publication are outside the solution, consistent with the baseline exclusions (M-0826).

Users. The baseline personas (BB personas) are consolidated into the roles that appear in the methodology: Owing Unit Representative (prepares and coordinates the case), BIDSC Reviewer (Benchmarking and Impact Assessment), Legal Reviewer, Strategy Reviewer, Finance Reviewer, and Decision Authority (screening, sufficiency and Decision Package control points). A deputyship-level overview for senior users is a P1 addition (M-0826). Role allocation at Gate 1, at option selection and for BIDSC execution is subject to open decisions OD-1 to OD-3.

External boundaries. Sanad supplies historical legislation as a reference source and retains statutory processing. Istitlaa is the public-consultation platform; exchange is manual. Maarifa holds internal documents and strategies and is reached through the Enterprise Knowledge Engine, with the connection route to be confirmed in Alpha. Saudi Open Data is the first source for Impact inputs, followed by user upload; internal permit data may be connected within the ministry boundary only.

3 Product principles

The following principles were agreed with the Solution Owner and apply to every module. Several make baseline capabilities concrete: the human-in-the-loop gateway (BB-14), explainable AI (BQ-05) and the advisory nature of outputs (BB exclusions). They are restated in individual requirements only where a module needs a specific rule.

ID	Principle	Rule	Source
PP-01	AI proposes first, humans decide	Each step opens with an AI-generated proposal. Human review follows the proposal; the workflow never waits for human input before AI has produced its suggestion.	BB-14, M-0818
PP-02	Comment, do not edit	Users do not edit AI-generated text directly. They comment on it, the system regenerates the content in response, and the user confirms the result.	M-0826
PP-03	Step-by-step confirmation	Every Policy Study content, option decision, draft state and feedback disposition requires an explicit human confirmation before it counts as complete. No step may be skipped.	BB-14, M-0826
PP-04	No comment ignored	Every reviewer or consultation comment receives a recorded disposition and a reason. A comment that cannot be used is recorded as such with the reason; it is never silently closed.	BB scope consultation synthesis, M-0818
PP-05	Forward-only feedback	Feedback produces the next version of the content or draft. Cases do not return to earlier stages because of feedback; material issues are resolved in the next version with a recorded reason.	M-0817, M-0818
PP-06	Every figure has a reference	Every number, ratio, range or forecast shown to users carries a source reference or an explicit assumption label. Retrieved results without a verifiable source are excluded.	BQ-05, M-0811, M-0819
PP-07	Local data first	Evidence is sourced from local Saudi data first, then international evidence adjusted with local factors, then explicitly labelled assumptions. The source tier is shown with the analysis. The process never blocks for missing data.	M-0819

PP-08	Alerts, not barriers	Screening and conflict checks present alerts and recommendations. The responsible human decides whether to proceed; the system does not hard-block on detected overlap.	BB-03, M-0824
PP-09	Advisory outputs, accountable roles	System recommendations are advisory. The Owning Unit is accountable for the case outputs; reviewers are accountable for their assessments; the Decision Authority is accountable for control-point decisions.	BB exclusions, M-0824
PP-10	Mechanisms, not topics	Benchmarking compares policy mechanisms and their conditions, not country or topic labels. Mechanism performance is rated independently of the case it came from.	BB scope benchmarking, M-0818, expert review 2 Sep

4 Functional requirements

4.1 Case intake and clarification

ID	Requirement	Priority	Phase	Source
PR-INT-01	Register a Policy Case with title, initial idea, objective, sector or affected area, initial evidence and known overlap. The objective and the intended approach are mandatory so that AI assistance can work from a stated goal.	P0	Alpha	BB-01, BB-13, M-0818, M-0824
PR-INT-02	Classify the legislative need by trigger type: Legislative Gap, Outdated Regulation, Rising Risk, Mandate Change. The classification is shown throughout the case.	P0	Alpha	M-0824, LDM step 1
PR-INT-03	Record the trigger source (systematic, direction from a higher authority, manual) in the case data model. Form registration defaults to manual; direction and systematic sources arrive through external channels. A direction-sourced case cannot be closed at screening; it may only proceed or be referred.	P1	Alpha	M-0817
PR-INT-04	Offer an AI clarification step after registration: the system proposes at least three structured clarification questions based on the topic, the user selects or completes answers,	P0	Alpha	M-0819

	and the system restates the policy question for confirmation. Questions, answers and versions are retained with the case.			
PR-INT-05	Check the stated objective against the ministry's mandate at registration and flag topics that appear to fall outside MOMAH's authority, with the reason and the likely responsible entity.	P1	Alpha	BB-03, M-0817, M-0811
PR-INT-06	Show the amendment path (update an existing instrument) as a visible but disabled option labelled as a later phase.	P1	Alpha	BB scope amendments, M-0826

4.2 Opportunity Screening (Gate 1)

ID	Requirement	Priority	Phase	Source
PR-SCR-01	Present a screening summary with the policy topic, initial evidence, possible duplication or conflict with existing instruments found in the knowledge base, similar past cases, and the value of further study.	P0	Alpha	BB-03, BB-04, M-0824
PR-SCR-02	Present detected overlap or conflict as an alert with an AI recommendation (proceed or close) and supporting evidence. The alert never blocks the decision.	P0	Alpha	BB-03, M-0824
PR-SCR-03	Record one of four outcomes with a mandatory reason: Proceed, Close, Park, Refer. Only Proceed opens the Policy Study. The recorded decision and reason remain visible in the case history.	P0	Alpha	BB-13, BB-14, LDM step 1
PR-SCR-04	Allow the user to challenge an AI screening finding and request re-assessment; the revised finding is shown with what changed.	P1	Alpha	BQ-05, M-0824
PR-SCR-05	Where data is incomplete, allow screening to continue on stated assumptions rather than block; assumptions are recorded with the decision.	P0	Alpha	M-0824

Open decision OD-1 determines whether the Owning Unit or the Decision Authority records the screening outcome.

4.3 Policy Study (nine contents)

ID	Requirement	Priority	Phase	Source
PR-PS-01	Provide the nine Policy Study contents as separate work items: 1 Gaps and Challenges; 2 Mandate and Legal Basis; 3 Alignment with Approved Strategies; 4 Value Chain and Responsibilities; 5 Stakeholders and Role Overlap; 6 Related Legislation (including instruments to be repealed or amended); 7 Local and International Benchmarks; 8 Expected Impact Analysis; 9 Implementation Budget.	P0	Alpha	BB scope concept builder, LDM step 2, M-0818
PR-PS-02	Generate an AI proposal for each content from the case record and the knowledge base. Proposals are labelled as AI-assisted and show their sources.	P0	Alpha	BB-02, BQ-05
PR-PS-03	Allow users to comment on any part of a proposal; the system regenerates the content in response to comments. Generated text is not directly editable. Comment history and regeneration versions are retained.	P0	Alpha	M-0826
PR-PS-04	Require explicit confirmation of each content by the Owning Unit. Show completion status for all nine contents on one overview.	P0	Alpha	BB-14, M-0826
PR-PS-05	Allow contents to be worked in parallel, subject to dependencies: content 8 uses the baseline and options from content 7; content 9 uses content 8. Dependency status is shown.	P0	Alpha	M-0816
PR-PS-06	Disable submission for internal review until all nine contents are confirmed, and list the incomplete items.	P0	Alpha	BB-13, M-0826
PR-PS-07	When a confirmed content is materially changed, clear its confirmation and the confirmations of dependent contents, and flag them for re-review.	P0	Alpha	BB-06 traceability, M-0824
PR-PS-08	Assemble the confirmed contents into an Initial Concept document (version V0) for review, with a change log for later versions.	P0	Alpha	BB scope policy memoranda, M-0817, M-0824
PR-PS-09	Provide a standalone "Find Gaps" research tool that runs the gap analysis outside a case, for exploratory use.	P2	Beta	BB-04, M-0826

Open decision OD-6 determines whether confirmation must be strictly sequential.

4.4 Benchmarking (content 7)

ID	Requirement	Priority	Phase	Source
PR-BM-01	Define the benchmark question from the confirmed policy question: target indicators, geography scope and time window. Parameter changes re-run the search and show what changed.	P0	Alpha	BB scope benchmarking, BM
PR-BM-02	Search local Saudi precedents first. Where no comparable local precedent passes the criteria, record that result and proceed to international cases.	P0	Alpha	M-0819
PR-BM-03	Apply four independently switchable eligibility gates to candidate cases (implemented in practice; verifiable source; policy instrument rather than generic risk management; within regulatory scope). Show the gate result and exclusion reason for every candidate.	P0	Alpha	BM, M-0818
PR-BM-04	Exclude any retrieved result that lacks a verifiable source.	P0	Alpha	BQ-05, M-0811
PR-BM-05	Build a case pool of at least three qualified cases and at least one cautionary case. Cautionary cases are shown separately and are not used as models to follow. Allow the user to nominate an additional candidate.	P0	Alpha	BM, M-0818
PR-BM-06	Score comparability with study-specific criteria and weights that total 100 percent. Show each criterion, weight, per-case score and the weighted result. The score is a similarity measure, not a probability of success.	P0	Alpha	BB scope "contextualized for Saudi economy", M-0818
PR-BM-07	Allow criteria to be relaxed transparently when they yield no result, with the relaxation recorded.	P1	Alpha	M-0811
PR-BM-08	Extract mechanism cards from qualified cases: instrument, conditions, observed outcome, source, evidence quality. Rate mechanism performance independently of the case it belongs to.	P0	Alpha	BM, expert review 2 Sep
PR-BM-09	Assess Saudi Transferability for each mechanism as applicable, applicable with adjustment, or not applicable, with reasons, required adjustments and relevant Saudi	P0	Alpha	BB scope benchmarking, M-0818

	legal constraints. Users accept or challenge each treatment individually or as a reviewed set.			
PR-BM-10	Construct Policy Options from accepted mechanisms. Mechanisms confirmed as not applicable are excluded. Each option shows its combined mechanisms, conditions, advantages, risks, sources and unresolved evidence. Each option carries three validation points: within MOMAH mandate; existing instrument covers it or not; new instrument or amendment.	P0	Alpha	BB-02, M-0816, M-0817
PR-BM-11	Present the local baseline used for comparison with observed inputs, cited sources, estimation method, units, time period and assumptions. Users can challenge an input and see the revised result. The same baseline is reused by Impact Assessment.	P0	Alpha	BB data sources data sources, M-0819
PR-BM-12	Produce a Benchmarking report for review that follows the six-step methodology and can be exported as part of the Policy Study.	P1	Alpha	M-0818

4.5 Impact Assessment (content 8)

The baseline Policy Impact Simulation capability (BB-09 to BB-11) is delivered as an ex-ante Impact Assessment of policy options against a baseline. The Solution Owner confirmed the five legally mandated assessment dimensions on 2 September; agent-based simulation, black-swan stress testing and probabilistic success scoring are recorded as deferred in Section 10.

ID	Requirement	Priority	Phase	Source
PR-IA-01	Define a Baseline Scenario (no intervention) projected over a stated horizon, separate from the Policy Options.	P0	Alpha	BB-10, M-0819
PR-IA-02	Assess each of the three Policy Options against the baseline across five dimensions: Economic, Financial, Employment, Social, Public Health. For each dimension the system first proposes whether it applies to the case; the user confirms or excludes it with a reason.	P0	Alpha	BB scope impact summaries, M-0819, M-0902
PR-IA-03	Produce a required-data list before analysis, showing which inputs are available locally, which come from international evidence with local adjustment, and which are assumptions. Users can upload missing data.	P0	Alpha	BB data sources, M-0819, M-0826

PR-IA-04	Label every input by source tier (local, international adjusted, assumption) and carry the label into results. Express international-derived figures as ranges, not single values.	P0	Alpha	BQ-05, M-0819
PR-IA-05	Select the calculation method per case from a method catalogue and show the chosen method, its assumptions and the calculation chain.	P1	Alpha	M-0819
PR-IA-06	Show effects by affected group (government, end consumers, private sector including SMEs, employment) and identify likely behavioural substitution or unintended effects.	P1	Alpha	BB scope risks and trade-offs, M-0819
PR-IA-07	Provide sensitivity information for the main assumptions and show uncertainty and limitations with every result.	P0	Alpha	BB scope sensitivity analysis
PR-IA-08	Where a reviewer disputes a figure, require the alternative value or its basis to be entered with the comment.	P1	Alpha	M-0818
PR-IA-09	Show an Ex-Post Evaluation Plan as a planning item only; the solution does not execute post-implementation evaluation.	P2	Beta	BB-15 deferred, LDM step 9

4.6 Stakeholder Review

ID	Requirement	Priority	Phase	Source
PR-SR-01	Route the confirmed Initial Concept to specialist reviewers in parallel: BIDSC (contents 7 and 8), Legal (contents 2 and 6), Strategy (content 3), Finance (contents 4 and 9). The mapping between the seven reviewing entities and the reviewer personas is configurable.	P0	Alpha	BB-14, BB responsibility matrix, M-0816, M-0817
PR-SR-02	Notify reviewers by email with a direct link to the assigned content; the review is tracked as a work item with status and due date.	P0	Alpha	M-0817
PR-SR-03	Reviewers can see all nine contents but may comment only on the contents within their responsibility.	P0	Alpha	BB-08 role-based access, M-0824

PR-SR-04	Each comment is typed as Requires Change, Consensus, or Propose New Option, and carries text and an optional attachment. A reviewer with no findings records No Comments explicitly. An unanswered review remains open.	P0	Alpha	M-0824
PR-SR-05	When the Owning Unit closes comment collection, the system re-analyses each comment and proposes a response with a confidence indicator and sources.	P0	Alpha	BB scope consultation synthesis, M-0818, M-0824
PR-SR-06	The Owning Unit records a disposition for every comment: Accept, Partially Accept, or Cannot Be Used, with a reason. Accepted changes regenerate the affected content and produce the next Initial Concept version.	P0	Alpha	M-0818
PR-SR-07	Show the review record in the case: reviewer, comment, AI response, disposition, reason, resulting content version.	P0	Alpha	BB-06, BB-14
PR-SR-08	Allow a reviewer to request a second opinion by forwarding an item to a named person by email, with the response recorded against the item.	P1	Beta	Solution Owner session 23 June, M-0817

4.7 Sufficiency review and option decision (Gate 2)

ID	Requirement	Priority	Phase	Source
PR-G2-01	Provide an Evidence and Options Sufficiency view consolidating completion, reviewer responses, evidence quality, assumptions, uncertainty and unresolved concerns.	P0	Alpha	BB-14, M-0816
PR-G2-02	Present all Policy Options together with the AI-recommended option and the reason for the recommendation, in a study-level view that shows the depth of the research behind each option.	P0	Alpha	BB-02, M-0816, M-0824
PR-G2-03	Record the decision: confirm the recommended option, select another option, return the case for focused revision, or reject all options and request regeneration. A manually added option requires a recorded justification.	P0	Alpha	BB-14, M-0811, M-0816
PR-G2-04	Record the Preferred Policy Option and its rationale, close Phase 1 of the case, and enable Drafting only when all nine	P0	Alpha	BB-13, M-0816

	confirmations, required reviews and the sufficiency decision are complete.			
PR-G2-05	Allow a justified exception to proceed with a documented gap, recorded with the approving role and reason.	P1	Alpha	BB-14 overrides

Open decision OD-2 determines which role confirms the Preferred Policy Option.

4.8 Drafting and legal quality

ID	Requirement	Prio rity	Phase	Source
PR-DR-01	Open Drafting with the Preferred Policy Option, its rationale and the supporting study evidence carried forward; option name, scope and objective fields are pre-filled.	P0	Alpha	BB-02, M-0824
PR-DR-02	Generate draft V0 for the selected instrument type (Regulation, Requirement, Rule) using the structural template for that type. Templates are derived from approved MOMAH instruments and maintained per type.	P0	Alpha	BB-02, M-0817, M-0824
PR-DR-03	Link each clause to the study evidence and option element it implements, so reviewers can see how the research shaped the wording.	P0	Alpha	BQ-05, M-0811
PR-DR-04	Users comment on clauses to trigger regeneration; generated text is not directly editable. Each regeneration creates a new draft version with a change summary.	P0	Alpha	M-0826
PR-DR-05	Compare any two draft versions with changed text highlighted and linked to the comment that caused the change.	P0	Alpha	BB scope versioning, BB-06
PR-DR-06	Present advisory Legal Conflict Analysis: related legislation, possible conflict, overlap or duplication, source context and proposed treatment including repeal or replacement of a conflicting provision. State the scan coverage and prompt the user to add related instruments from other entities before running the analysis.	P0	Alpha	BB-03, M-0811, Solution Owner session 23 June
PR-DR-07	Provide the 39-item pre-consultation checklist with status, supporting evidence, reviewer observations and reasons for non-applicability. The checklist is completed by the	P0	Alpha	LDM step 3, M-0817

	Owning Unit and Legal Reviewer before the draft is marked Consultation-Ready.			
PR-DR-08	Run the internal review of draft V0 by the reviewing entities using the Stakeholder Review mechanism (Section 4.6), producing draft V1.	P0	Alpha	BB-14, M-0817
PR-DR-09	Mark the draft Consultation-Ready when the checklist and internal review are complete. This status carries no legal effect.	P0	Alpha	BB exclusions, LDM step 4
PR-DR-10	Prepare a dissemination plan and a translated version of the draft as supporting outputs.	P3	Later	M-0811

4.9 Consultation feedback

ID	Requirement	Priority	Phase	Source
PR-CF-01	Record the consultation as a case stage: consultation material prepared, manual submission to Istitlaa recorded with date and reference, consultation window, and feedback received. No system connection to Istitlaa.	P0	Alpha	BB scope consultation synthesis, M-0817
PR-CF-02	Import consultation comments by manual entry or file upload; the system proposes a classification (minor or material) and a summary of themes as advisory input.	P0	Alpha	BB scope theme clustering, M-0817
PR-CF-03	Record a disposition for every comment: Accepted, Partially Accepted, or Rejected, with a reason. The consultation stage cannot close while any comment lacks a disposition.	P0	Alpha	M-0818
PR-CF-04	Apply accepted comments to produce draft V2 through regeneration; the case does not return to option assessment or the Policy Study.	P0	Alpha	BB scope amendment options, M-0817
PR-CF-05	Link each accepted comment to the clause change it produced.	P0	Alpha	BB-06 traceability

Open decision OD-7 confirms whether the consultation stage is within the current delivery or a later phase.

4.10 Review-Ready outputs and Decision Package

ID	Requirement	Priority	Phase	Source
PR-DP-01	Assemble the Decision Package: Policy Study summary, Benchmarking findings, option comparison, Impact Assessment, Preferred Policy Option rationale, review record, consultation dispositions, draft status, assumptions, limitations and risks.	P0	Alpha	BB scope policy memoranda, BB-14
PR-DP-02	Label the final draft Review-Ready Regulatory Draft. The label states that the draft is not approved, published or legally effective.	P0	Alpha	BB exclusions
PR-DP-03	Present the Decision Package to the Decision Authority for review with the recorded decision (accept for downstream processing, or return with reason).	P0	Alpha	BB-14, M-0826
PR-DP-04	Export the Policy Study report, Benchmarking report and Decision Package as documents for downstream statutory processing.	P1	Beta	BB integrations DMS

4.11 Knowledge base and Policy Library

ID	Requirement	Priority	Phase	Source
PR-KB-01	Hold current instruments from the Istitlaa platform and the MOMAH website (approximately 200 items collected), the Council of Ministers decisions reference, and instruments from REGA, REDF, SAMA and MOF, each with source, version, date, legal status and issuing entity.	P0	Alpha	BB-01, BB-06, BB data sources, M-0824, M-0826
PR-KB-02	Where an approved version exists, treat it as authoritative over draft versions; retain draft versions with a Draft or Consultation status label.	P0	Alpha	M-0826
PR-KB-03	Layer the knowledge base: Law (authorising layer); Regulation, Requirement and Rule (mandatory instruments); Guide, strategy documents, historical drafts and international cases (supporting context).	P1	Alpha	BB-01 policy object, M-0817

PR-KB-04	Refresh Istitlaa and MOMAH website sources on a schedule and flag new or changed instruments.	P1	Beta	M-0826
PR-KB-05	Retrieve permitted internal reference material, including Maarifa-origin documents and strategies, through the Enterprise Knowledge Engine. Retrieved evidence is shown separately from generated analysis.	P0	Alpha	BB data sources legal repository, M-0818
PR-KB-06	Retrieve historical legislation from Sanad as a reference source. Sanad performs no workflow step.	P1	Beta	M-0816
PR-KB-07	Provide Policy Library browse and search by title, type, entity, status and topic, with the original source and relationships to related instruments.	P0	Alpha	BB-06, BB scope registry
PR-KB-08	Use the same knowledge base for Gate 1 duplication and conflict checks, Benchmarking local precedent search and Legal Conflict Analysis.	P0	Alpha	BB-03, M-0826

4.12 Workspace, dashboard and administration

ID	Requirement	Priority	Phase	Source
PR-WS-01	Provide a case workspace showing topic, lifecycle status, responsible roles, evidence, review activity, decisions and the actions available to the current persona.	P0	Alpha	BB-08, BB-13
PR-WS-02	Provide a dashboard of cases by stage, pending reviews and pending decisions, with metric definitions available on hover.	P1	Alpha	BB-13, Solution Owner session 23 June
PR-WS-03	Provide a deputyship-level overview for senior users showing case portfolio and status without editing rights.	P1	Beta	BB personas senior leaders, M-0826
PR-WS-04	Maintain a full case history: every AI proposal, comment, regeneration, confirmation, decision and reason with actor and timestamp.	P0	Alpha	BB-06, BB-14

PR-WS-05	Show the wider legislative lifecycle as context, with the Policy AI Lab boundary marked, without changing statutory responsibilities.	P1	Alpha	LDM, M-0811
PR-WS-06	Provide user preferences limited to language and interface settings. Knowledge ingestion, indexing, user and permission administration are administrative functions outside the end-user interface.	P1	Alpha	BB-08

4.13 Roles and access

ID	Requirement	Priority	Phase	Source
PR-RA-01	Support the personas Owning Unit Representative, BIDSC Reviewer, Legal Reviewer, Strategy Reviewer, Finance Reviewer and Decision Authority, each with a role-scoped view and action set.	P0	Alpha	BB personas, BB responsibility matrix, M-0816
PR-RA-02	Represent the Owning Unit by a single named representative per case; working-group structures are not modelled.	P0	Alpha	M-0816
PR-RA-03	Authenticate users through MOMAH/NHCI SSO and derive organisational identity from ASAS in the operational solution. Access to case information does not by itself confer editing or decision rights.	P0	Beta	BB integrations, BQ-01, DSR
PR-RA-04	Apply role-based data access so that restricted internal data (for example permit data) is visible only to authorised roles within the ministry boundary.	P0	Beta	BB-08, M-0826, Solution Owner session 8 July

4.14 Language and layout

ID	Requirement	Priority	Phase	Source
PR-LG-01	Provide the interface in English and Arabic with right-to-left layout for Arabic; switching preserves the current case and step.	P0	Alpha	BQ-04

PR-LG-02	Generate drafts and study content in Arabic as the authoritative language, with English available for review; policy and legal terminology receives human review before release.	P0	Beta	BQ-04, M-0902
PR-LG-03	Support desktop as the primary environment with usable tablet and mobile layouts for the principal workflow.	P0	Alpha	BB-08

5 Non-functional requirements

The non-functional requirements start from the baseline quality attributes BQ-01 to BQ-06 (security, performance, scalability, language, explainable AI, sovereign compliance) and add the product-level detail agreed with the Solution Owner. The baseline reference is given for each.

ID	Category	Requirement	Baseline ref
PN-01	Performance	Standard page and API requests meet a three-second P95 target under normal load. Generation, retrieval and analysis tasks that take longer show progress and end in a clear completion or failure state; initial draft generation completes within 60 seconds at P95. Prototype: navigation and predefined state changes within two seconds for 95 percent of actions.	BQ-02
PN-02	Security and data protection	Approved identity services, role-based access control and least-privilege permissions; encryption in transit and at rest; auditable access to restricted content and material changes. Restricted internal data does not leave the ministry boundary. Prototype contains no production credentials, personal data or live administrative data.	BQ-01
PN-03	AI and retrieval quality	Before operational acceptance, capabilities are evaluated on representative Arabic and English cases for retrieval relevance, citation completeness and correctness, evidence grounding, legal-conflict precision and recall, bilingual output quality and unsupported-claim rate. Critical legal or citation errors block release. Model selection is validated by comparing candidate models on the same cases with a cost comparison (M-0818). Predefined Prototype outputs are not evidence of model accuracy.	BQ-05
PN-04	Devices and platforms	Desktop Web primary; current Chrome, Edge and Safari; usable tablet and mobile layouts for the principal workflow.	BB-08

PN-05	Language and layout	English and Arabic interface; Arabic right-to-left; human language review of policy and legal terminology; switching preserves context.	BQ-04
PN-06	Availability	At least 95 percent in the first six months, 98 percent in Year 1 and 99 percent in Year 2, measured during KSA business hours.	New
PN-07	Evidence transparency	Every figure, ratio, range or forecast carries a source reference or an explicit assumption label. Retrieved results without a verifiable source are excluded. The source tier (local, international adjusted, assumption) is shown wherever the figure appears.	BQ-05
PN-08	Traceability and audit	Every AI proposal, comment, regeneration, confirmation, disposition and decision is retained with actor, timestamp and reason, and can be reconstructed for a case at any later date.	BB-06, BB-14
PN-09	Human accountability	Screening, sufficiency, option, review and feedback decisions require an explicit human action by an authorised role. System-generated content is labelled advisory and is not directly editable.	BB-14, BB exclusions
PN-10	Data residency and model hosting	AI inference and data storage remain within KSA under applicable PDPL and NCA controls; internal documents are not sent to external models. Public international research may use external models where approved.	BQ-06
PN-11	Scalability	The knowledge base and case store scale with the growth of collected instruments and concurrent cases; sizing assumptions for users and parallel cases are stated in the Alpha design.	BQ-03

6 Data and integration requirements

The business baseline lists the target integrations and data sources. The table below records the current position for each after the Solution Owner sessions and the 20 July data and integration support request.

ID	Source or system	Requirement	Phase	Source
DI-01	Istitlaa and MOMAH website	Current instruments collected and refreshed on a schedule; approved version authoritative, drafts retained with status.	Alpha	M-0824, M-0826

DI-02	REGA, REDF, SAMA, MOF	Approved instruments collected from each entity's official site into the knowledge base.	Alpha	BB data sources, M-0826
DI-03	Council of Ministers decisions	Reference file supplied by the Solution Owner loaded into the knowledge base; item count to be confirmed.	Alpha	Solution Owner, September
DI-04	Maarifa via Enterprise Knowledge Engine	Internal documents and strategies retrievable for research; connection route and access scope confirmed in Alpha.	Alpha	BB data sources legal repository, M-0818
DI-05	Sanad	Historical legislation retrievable as reference; no workflow handoff.	Beta	DSR, M-0816
DI-06	Saudi Open Data	First source for Impact inputs; datasets filtered by organisation; API access where available.	Alpha	M-0826
DI-07	User upload	Structured data upload for Impact inputs not available from public sources, with source and date recorded.	Alpha	M-0826
DI-10	International data (World Bank, OECD, IMF)	Accessed by API or document extraction for benchmarking and impact evidence; every item source-verified.	Alpha	BB data sources, M-0811
DI-11	Identity (MOMAH/NHCI SSO, ASAS, Nafath)	SSO for authentication; ASAS for organisational identity; Nafath only if a formal signing step is later brought into scope.	Beta	BB integrations, DSR

7 Out of scope and deferred

The baseline exclusions are carried forward unchanged. The Solution Owner sessions added the deferrals listed after them.

- Automated policy decisions, political prioritisation, authoritative legal advice, or replacement of accountable reviewers (BB exclusions).
- Statutory approval, electronic signature, official publication and issuance of legally effective instruments; direct submission to legislative bodies or gazette systems (BB exclusions, M-0826).

- National macroeconomic forecasting platform, real-time compliance monitoring, regulatory enforcement, municipal service operations, citizen-facing applications, foundation model development (BB exclusions).
- Workflows to amend, replace, consolidate or repeal existing instruments (deferred to a later phase; placeholder shown). Guide authoring and Law drafting (M-0817, M-0826).
- System connection to Istitlaa; operation or replacement of external consultation, publication or government records platforms (M-0817).
- Post-implementation quality measurement and continuous model retraining from real-world outcomes (BB-15; later phase).
- Native mobile or tablet applications.
- The central-government approval path for entirely new policy ideas (retained as a placeholder in the lifecycle view only) (M-0811).

8 Prototype boundary

The Prototype demonstrates the requirements above on one representative municipal policy case using public information and predefined records. It runs as a front-end application with deterministic states and a reset to the initial state, a persona switcher in place of production identity, no live AI model, no persistent storage, no working file upload or export, and no connection to Sanad, Istitlaa or Maarifa. Comment-triggered regeneration, AI re-analysis of review comments, and challenge-and-revise interactions are shown through predefined alternative states. The Prototype knowledge base is a preloaded subset of the collected instruments with approved and draft status labels. Prototype review assesses the workflow, role responsibilities, evidence presentation and bilingual behaviour; it does not measure model performance or KPIs.

9 Open decisions

ID	Decision needed	Current design position	Owner
OD-1	Which role records the Gate 1 screening outcome: Owing Unit (per M-0824) or a separate Decision Authority.	Decision Authority, as a Prototype simplification.	Solution Owner
OD-2	Which role confirms the Preferred Policy Option: Owing Unit, or the Gate 2 decision authority (per M-0811, M-0816).	Owing Unit selects after the sufficiency decision.	Solution Owner

OD-3	Whether BIDSC prepares Benchmarking and Impact Assessment with the Owning Unit (per M-0811, M-0816) or reviews them only.	BIDSC reviews; Owning Unit prepares.	Solution Owner
OD-4	Whether the Policy Study distinguishes Policy from Regulation outputs at the start of the study.	Not answered at M-0818 and M-0824; instrument type is selected at Drafting.	Solution Owner
OD-5	Label for a comment that is not adopted: Rejected or Cannot Be Used.	Cannot Be Used for internal review (M-0818); Rejected for consultation.	Solution Owner
OD-6	Whether the nine contents may be confirmed in parallel or must be confirmed in sequence.	Parallel work with dependency control (PR-PS-05).	Solution Owner
OD-7	Whether the consultation feedback stage (draft V1 to V2) is part of the current delivery.	Included at P0; M-0817 did not close the point.	Solution Owner
OD-8	Whether Law is absolutely excluded as an output.	Law production excluded; Law is a knowledge-base layer.	Solution Owner
OD-9	Connection route to Maarifa: direct API or through the Enterprise Knowledge Engine.	Enterprise Knowledge Engine pending Alpha integration design.	ByteDance with Oversight Team
OD-10	Which Open Data platform is authoritative for Impact inputs and whether API access is provided.	National Saudi Open Data platform.	Solution Owner

10 Relationship to the business baseline

The table reconciles this document with the business baseline. Carried means the baseline capability is delivered as first described; Refined means the Solution Owner sessions changed how it is delivered, with the change noted; Deferred means the sessions placed it in a later phase. Where the two differ, the Solution Owner decision governs.

BB	Feature	Status	Product requirements and notes
BB-01	Policy Entity	Carried	PR-INT-01, PR-KB-01, PR-KB-03. Policy Case and knowledge-base instrument are the two entity types.
		Refined	

BB-02	Intelligent Policy Generator and Synthesizer		PR-PS-02, PR-BM-10, PR-G2-02, PR-DR-01 to PR-DR-05. Generation is split into Policy Study proposals, option construction and draft generation; drafting follows the option decision (M-0811); comment-driven regeneration replaces direct editing (M-0826).
BB-03	Regulatory Conflict Engine	Carried	PR-SCR-01, PR-SCR-02, PR-DR-06, PR-KB-08. Advisory alerts at screening and at drafting; scan coverage stated.
BB-04	Policy Gap and Opportunity Analysis	Refined	PR-PS-01 content 1, PR-SCR-01, PR-PS-09. Gap analysis is content 1 of the Policy Study; standalone tool at P2.
BB-05	Vision 2030 Alignment Auditor	Refined	PR-PS-01 content 3. Alignment with approved strategies is reviewed by the Strategy Reviewer; ministry-strategy alignment added alongside Vision 2030; no numeric alignment score.
BB-06	Institutional Memory Vault	Carried	PR-KB-01 to PR-KB-07, PR-WS-04, PN-08.
BB-07	Regional Impact Visualization	Deferred	Regional breakdown may appear within Impact Assessment results where data allows; geospatial heatmaps are not in the current delivery.
BB-08	Web-based role-based platform	Carried	PR-WS-01, PR-RA-01 to PR-RA-04, PR-LG-03.
BB-09	Black-Swan Policy Simulation	Deferred	Stress scenarios are not part of the ex-ante Impact Assessment; sensitivity on main assumptions is provided (PR-IA-07).
BB-10	Multi-Variable Simulation Lab (ABM)	Refined	PR-IA-01 to PR-IA-07. Delivered as ex-ante Impact Assessment of options against a baseline across five mandated dimensions with labelled data tiers; agent-based modelling is not used (Solution Owner sessions June, M-0819, M-0902).
BB-11	Predictive Success Scoring	Deferred	No probabilistic success score or automatic fail threshold; comparability scores in Benchmarking are similarity measures only (PR-BM-06).
BB-12	Risk Escalation	Refined	Risks and unresolved concerns are surfaced in the sufficiency view and Decision Package (PR-G2-01, PR-DP-01); automated escalation tiers are a later phase.
BB-13	Policy Lifecycle Orchestrator	Refined	PR-INT-01, PR-SCR-03, PR-PS-06, PR-G2-04, PR-WS-01. Stages follow the Legislative Development Methodology: screening, Policy Study, review,

			sufficiency and option, drafting, checklist, consultation, Decision Package; approval and enactment are outside the solution.
BB-14	Human-in-the-Loop Gateway	Carried	PP-01 to PP-05, PR-SR-01 to PR-SR-08, PR-G2-01 to PR-G2-05, PR-DP-03, PN-09. Every control point is a recorded human decision; overrides require a justification.
BB-15	Continuous Learning Loop	Deferred	Post-implementation outcome data and model retraining are outside the current delivery; an Ex-Post Evaluation Plan is shown as a planning item (PR-IA-09).